

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0509

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

- (a) Explain the scenario and characteristics of this noise.
- (b) Identify key issues impacting image quality.
- (c) Apply an appropriate filtering technique.
- (d) Justify why this method is more suitable than alternatives.
- (e) Present your response clearly and logically.

ANSWER:

(a) Explain the Scenario and Characteristics

Salt-and-pepper noise is a type of **impulse noise** that appears as random black (pepper) and white (salt) pixels in an image. It usually occurs due to errors during image acquisition, transmission, scanning, or faulty camera sensors. In a scanned document, this noise affects readability by covering letters and symbols

with unwanted pixels. As the amount of noise increases, the document becomes difficult to read and process using Optical Character Recognition (OCR).

Characteristics

- Appears as random black and white pixels.
- Common in scanned documents and transmitted images.
- Changes only a few pixels while leaving others unchanged.
- Reduces image quality and readability.

(b) Identify Key Issues

The major issues caused by salt-and-pepper noise include:

- Poor readability of printed text.
- Loss of important document information.
- Difficulty in recognizing characters using OCR.
- Reduced image quality.
- Incorrect segmentation and feature extraction.
- Increased processing time for image analysis.

(c) Apply an Appropriate Filtering Technique

The **Median Filter** is the best technique for removing salt-and-pepper noise.

Working

- Select a small neighbourhood around each pixel.
- Arrange neighbouring pixel values in ascending order.
- Replace the centre pixel with the median value.
- Repeat this process for every pixel.

This removes noisy pixels while preserving the original edges of characters.

(d) Justification

The Median Filter is preferred because it effectively removes impulse noise without blurring important edges. Unlike the Mean Filter, which averages neighbouring pixels and causes image blur, the Median Filter preserves sharp

text boundaries. Therefore, it provides better readability and higher OCR accuracy.

(e) Conclusion

The Median Filter is the most suitable method for restoring scanned documents affected by salt-and-pepper noise because it removes unwanted pixels while maintaining image details and text clarity.

2. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

- (a) Interpret the scenario and explain the importance of boundary detection.
- (b) Identify key challenges in detecting boundaries.
- (c) Apply a suitable edge detection method.
- (d) Justify your selection with logical reasoning.
- (e) Present your explanation clearly and systematically.

ANSWER:

(a) Interpret the Scenario

In medical imaging, identifying the boundaries between tissues, organs, or tumours is essential for diagnosis and treatment planning. Accurate boundary detection helps doctors measure organ size, detect abnormalities, and perform surgical planning. If boundaries are not detected correctly, diagnosis may become inaccurate.

(b) Key Challenges

Some common challenges include:

- Low contrast between tissues.
- Presence of image noise.
- Blurred or weak edges.
- Overlapping structures.

- Uneven illumination.
- Complex anatomical shapes.

(c) Apply a Suitable Edge Detection Method

The **Canny Edge Detection Algorithm** is the most suitable method.

Steps

1. Apply Gaussian smoothing to remove noise.
2. Calculate intensity gradients.
3. Perform non-maximum suppression.
4. Apply double thresholding.
5. Track edges using hysteresis.

The final output contains clear and continuous object boundaries.

(d) Justification

Canny Edge Detection provides high detection accuracy, reduces false edges, and produces thin, continuous boundaries. Compared to Sobel or Prewitt operators, Canny performs better in noisy medical images because it combines noise reduction with edge detection.

(e) Conclusion

Canny Edge Detection is widely used in medical imaging because it accurately identifies tissue boundaries while minimizing noise and false edge detection.

3. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

- (a) Interpret the scenario and explain segmentation requirements.
- (b) Identify key features enabling separation.
- (c) Apply a suitable segmentation technique.
- (d) Justify your decision logically.
- (e) Present your response clearly.

ANSWER:

(a) Interpret the Scenario

Image segmentation is the process of dividing an image into meaningful regions. In this scenario, the object is clearly distinguishable from the background because both have different intensity values. Therefore, a simple segmentation technique can separate the object effectively.

(b) Key Features

The important features include:

- Large intensity difference.
- Uniform background.
- Clearly visible object.
- High contrast.
- Minimal noise.

(c) Apply a Suitable Segmentation Technique

Threshold Segmentation

Working

- Choose an appropriate threshold value.
- Pixels with intensity greater than the threshold are classified as the object.
- Pixels with intensity below the threshold become the background.

This creates a binary image where the object is clearly separated.

(d) Justification

Thresholding is computationally simple, fast, and highly accurate when foreground and background have distinct intensity values. It requires less processing time compared to clustering or region-growing algorithms.

(e) Conclusion

Thresholding is the most efficient segmentation technique for images with clear intensity differences because it provides accurate object separation with low computational cost.

4. Low Brightness Image

An image captured in low light appears very dark with poor visibility.

- (a) Interpret the scenario and explain the cause of low brightness.
- (b) Identify key issues affecting image visibility.
- (c) Apply an appropriate enhancement technique.
- (d) Justify your choice with reasoning. (e) Present your explanation clearly.

ANSWER:

(a) Interpret the Scenario

Images captured under poor lighting conditions often appear dark, making it difficult to observe important objects and details. Low brightness may occur because of insufficient illumination, incorrect camera exposure, or environmental conditions such as night-time photography.

(b) Key Issues

The major problems include:

- Poor visibility.
- Low image contrast.
- Hidden object details.
- Difficulty in object detection.
- Reduced recognition accuracy.
- Poor overall image quality.

(c) Apply an Enhancement Technique

Histogram Equalization

Working

Histogram Equalization redistributes pixel intensity values over the entire intensity range, increasing image contrast and brightness.

(d) Justification

Histogram Equalization enhances both brightness and contrast automatically. It reveals hidden details without changing the actual image content. It is more effective than simply increasing brightness because it improves contrast across the entire image.

(e) Conclusion

Histogram Equalization is an effective enhancement technique for dark images because it improves visibility, increases contrast, and reveals hidden image details.

5. Gaussian Noise

An image contains Gaussian noise affecting smooth regions.

- (a) Interpret the scenario and describe Gaussian noise.
- (b) Identify key issues affecting quality.
- (c) Apply an appropriate smoothing filter.
- (d) Justify your method over alternatives.
- (e) Present your explanation clearly.

ANSWER:

(a) Interpret the Scenario

Gaussian noise is a random noise that follows a normal (Gaussian) distribution. It usually occurs because of electronic sensor noise, thermal effects, or low-light image acquisition. The noise spreads throughout the image and mainly affects smooth regions.

(b) Key Issues

The major issues include:

- Grainy appearance.
- Reduced image quality.
- Difficulty in feature extraction.
- Reduced segmentation accuracy.
- Blurred visual appearance.

(c) Apply an Appropriate Smoothing Filter

Gaussian Filter

Working

The Gaussian Filter applies weighted averaging using a Gaussian kernel. Nearby pixels contribute more to the output than distant pixels, resulting in smooth noise reduction.

(d) Justification

The Gaussian Filter is specifically designed to remove Gaussian noise while preserving the overall image structure. Unlike the Mean Filter, it performs smoother averaging, and unlike the Median Filter, it is better suited for normally distributed noise.

(e) Conclusion

The Gaussian Filter effectively reduces Gaussian noise and improves image quality while preserving important image structures.

6. Loss of Fine Details

An image loses fine details after processing.

- (a) Interpret the scenario and explain detail loss.
- (b) Identify key issues related to filtering.
- (c) Apply a suitable enhancement technique.
- (d) Justify your method with reasoning. (e) Present your explanation clearly.

ANSWER:

(a) Interpret the Scenario

Sometimes, after applying smoothing filters, small image details such as textures, edges, and thin lines become blurred or disappear. This loss of detail reduces image sharpness and affects feature extraction.

(b) Key Issues

- Loss of edges.

- Blurred textures.
- Reduced sharpness.
- Poor object recognition.
- Lower segmentation accuracy.

(c) Apply a Suitable Enhancement Technique

Unsharp Masking (Image Sharpening)

Working

- Create a blurred version of the image.
- Subtract it from the original image.
- Add the difference back to the original image.
- This enhances high-frequency details and sharpens edges.

(d) Justification

Unsharp Masking restores edge information without significantly increasing noise. It improves image sharpness more effectively than simple sharpening filters and preserves visual quality.

(e) Conclusion

Unsharp Masking is an excellent enhancement technique for restoring fine image details and improving overall sharpness.

7. Edge Enhancement Requirement

An application requires highlighting edges for feature extraction.

- (a) Interpret the scenario and explain the need for edge enhancement.
- (b) Identify key issues in detecting edges.
- (c) Apply an appropriate method.
- (d) Justify your choice logically.
- (e) Present your explanation clearly.

ANSWER:

(a) Interpret the Scenario

Many computer vision applications such as face recognition, fingerprint analysis, object detection, and medical imaging require clear edge information for feature extraction. Edge enhancement increases the visibility of object boundaries, making image analysis more accurate.

(b) Key Issues

- Weak edge contrast.
- Noise affecting edge detection.
- Missing boundary information.
- False edge generation.
- Poor feature extraction.

(c) Apply an Appropriate Method

Sobel Edge Detection

Working

The Sobel operator calculates horizontal and vertical gradients using two convolution masks. These gradients are combined to produce the edge magnitude image, highlighting object boundaries.

(d) Justification

The Sobel operator is simple, computationally efficient, and effective for detecting strong edges. It also provides gradient direction information, making it suitable for many real-time computer vision applications.

(e) Conclusion

The Sobel Edge Detection method effectively enhances image boundaries, improves feature extraction, and supports accurate object recognition in computer vision systems.

Code of Conduct:

I **N.AKSHAYA(192425129)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues

Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand